

## Introduction

The Bernoulli distribution models a single binary trial: success vs. failure, disease vs. healthy, click vs. no click. It is the simplest non-trivial distribution and the building block of the binomial, geometric, and negative binomial distributions.

## Prerequisites

Random variables and basic probability.

## Theory

A random variable  $X$  is **Bernoulli**( $p$ ) if it takes values in  $\{0, 1\}$  with

$$P(X = 1) = p, \quad P(X = 0) = 1 - p.$$

PMF:  $p_X(x) = p^x(1 - p)^{1-x}$  for  $x \in \{0, 1\}$ .

Moments:

- $E[X] = p$ .
- $\text{Var}(X) = p(1 - p)$ , maximised at  $p = 0.5$  (value 0.25).
- $E[X^k] = p$  for every  $k \geq 1$  (since  $X^k = X$ ).

The sum of  $n$  iid Bernoulli( $p$ ) variables is **Binomial**( $n, p$ ).

MLE of  $p$  from an iid sample  $X_1, \dots, X_n$ :  $\hat{p} = \bar{X}$ , the sample proportion.

## Assumptions

Independence across trials and a common probability  $p$ . Violations of either produce overdispersion (if  $p$  varies) or correlation (if trials are dependent).

## R Implementation

```
set.seed(2026)

p <- 0.3
n <- 10000

# Sample from Bernoulli(p) via rbinom with size = 1
X <- rbinom(n, size = 1, prob = p)

# Sample mean is the MLE of p
c(p_hat = mean(X), p_true = p)

# Sample variance approximates p(1 - p)
c(sample_var = var(X), theoretical = p * (1 - p))

# 95% Wald confidence interval for p
se_p <- sqrt(mean(X) * (1 - mean(X)) / n)
c(lower = mean(X) - 1.96 * se_p, upper = mean(X) + 1.96 * se_p)

# Wilson (better for small p or small n)
binom::binom.confint(sum(X), n, conf.level = 0.95, methods = "wilson")
```

## Output & Results

```
p_hat    p_true
0.3002    0.3000

sample_var theoretical
0.2101    0.2100

lower    upper
0.2912    0.3092

method    x    n    mean    lower    upper
Wilson 3002 10000 0.3002 0.2912 0.3094
```

Empirical mean is  $0.3002 \approx p$ ; empirical variance is  $0.2101 \approx p(1 - p) = 0.21$ ; Wald and Wilson CIs agree in this range.

## Interpretation

In applied work, the Bernoulli is the underlying distribution of every binary outcome: adverse event, responder status, screen-positive. Reporting usually aggregates to proportions across  $n$  independent Bernoullis – i.e., to the binomial framework.

## Practical Tips

- For small samples or extreme  $p$  (near 0 or 1), use Wilson or Clopper-Pearson CIs instead of Wald.
- Overdispersion (variance exceeding  $p(1 - p)$ ) in grouped data signals unobserved heterogeneity; consider beta-binomial or mixed-effects logistic regression.
- `rbinom(n, size = 1, prob = p)` is the direct way to simulate Bernoullis; `sample(0:1, n, replace = TRUE, prob = c(1 - p, p))` is equivalent.
- The Bernoulli likelihood is the foundation of logistic regression; MLE of  $p$  from regression coefficients requires the logit link.
- Two dependent Bernoullis require a specification of their joint distribution – usually via a copula or a 2x2 joint PMF.

## For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots